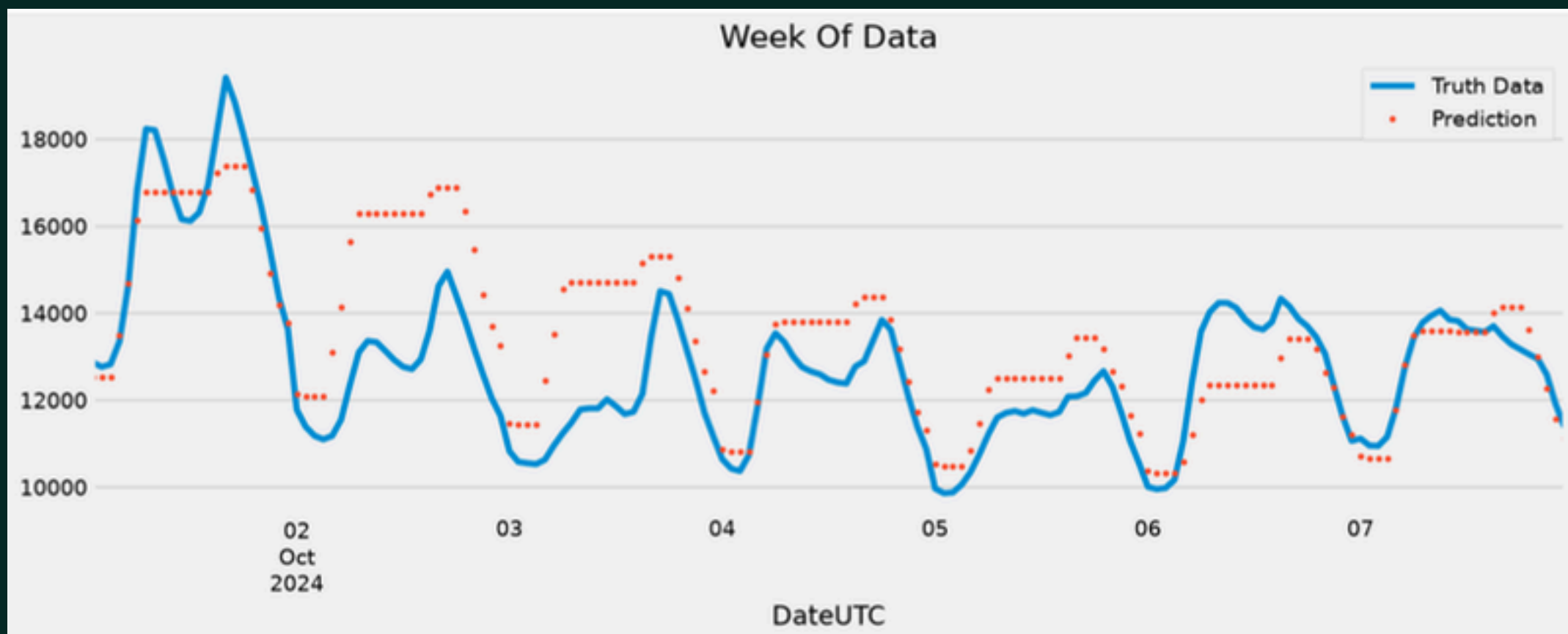
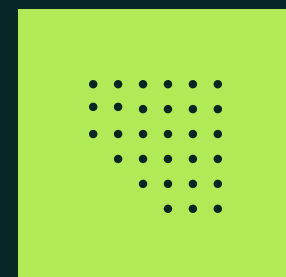




ENTSO-E Data



 **Mohamed Tawhid**
Energy Analyst | Power System Specialist

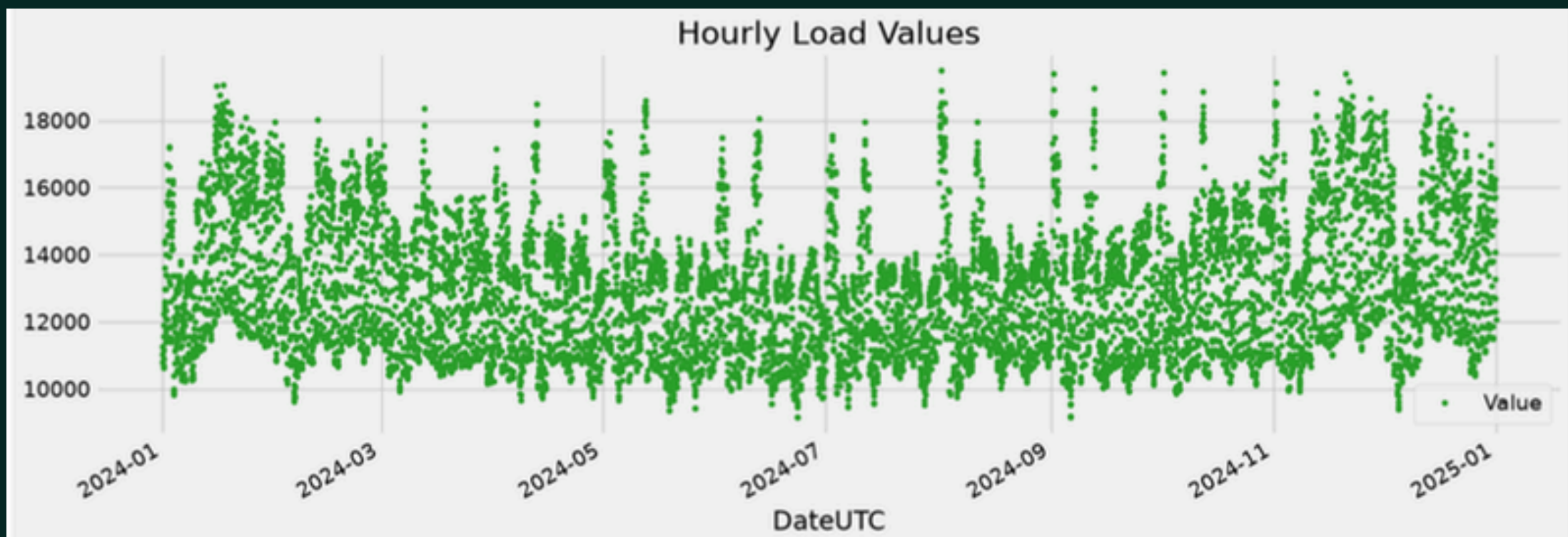
Time Series Forecasting for
ENTSO-e 2024 Energy Data

Forecasting The Netherlands' Energy
Demand Time Series Analysis



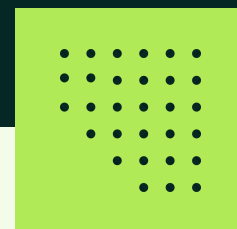


The Challenge: Predicting Complex Energy Demand



Hourly Load Values

Energy Distribution



Objective: To build a robust forecasting model for the Netherlands' hourly energy consumption to aid in grid management and capacity planning.

Dataset: ENTSO-e Transparency Platform

Scope: Full Calendar Year 2024 | 8,760 Hours | Country Code: NL

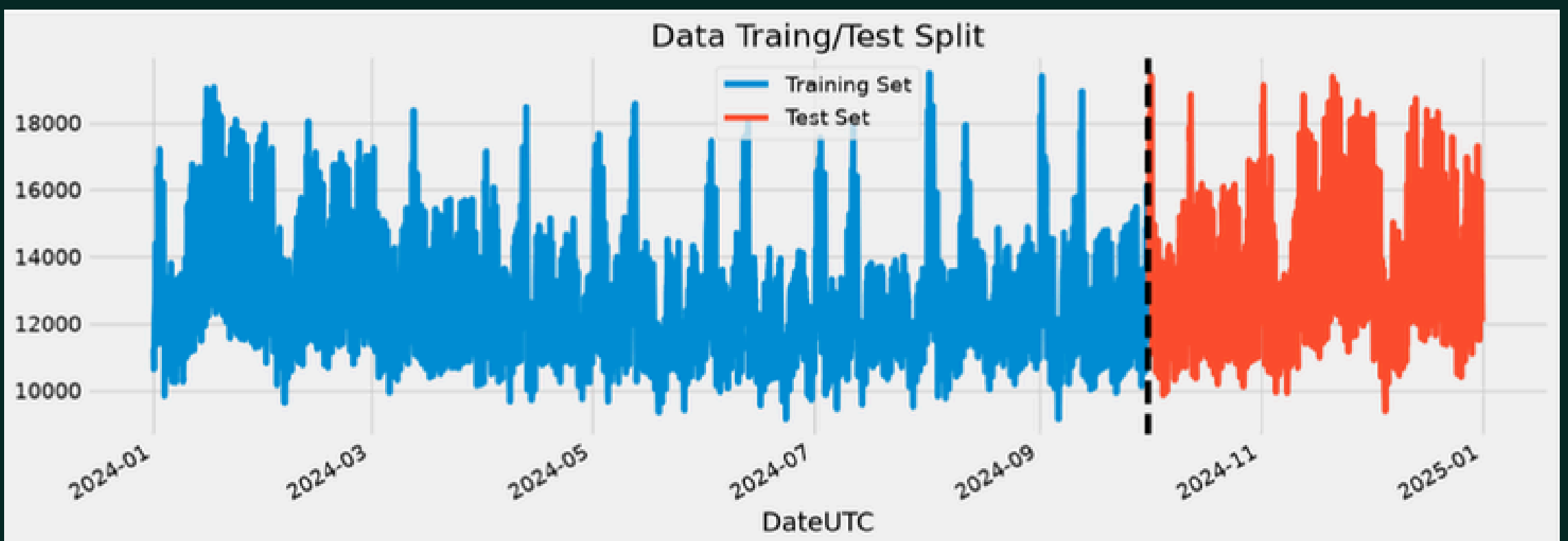
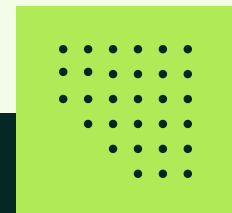


● ● ● Feature Engineering: Teaching the Model About Time

Transforming Time into Predictive Features

To help the ML model understand temporal patterns, we engineered features like hour of day, day of week, month, and day of year from the timestamp.

Key Takeaway: Feature engineering is crucial for unlocking the predictive power of time series data.



Data Traing/Test Split



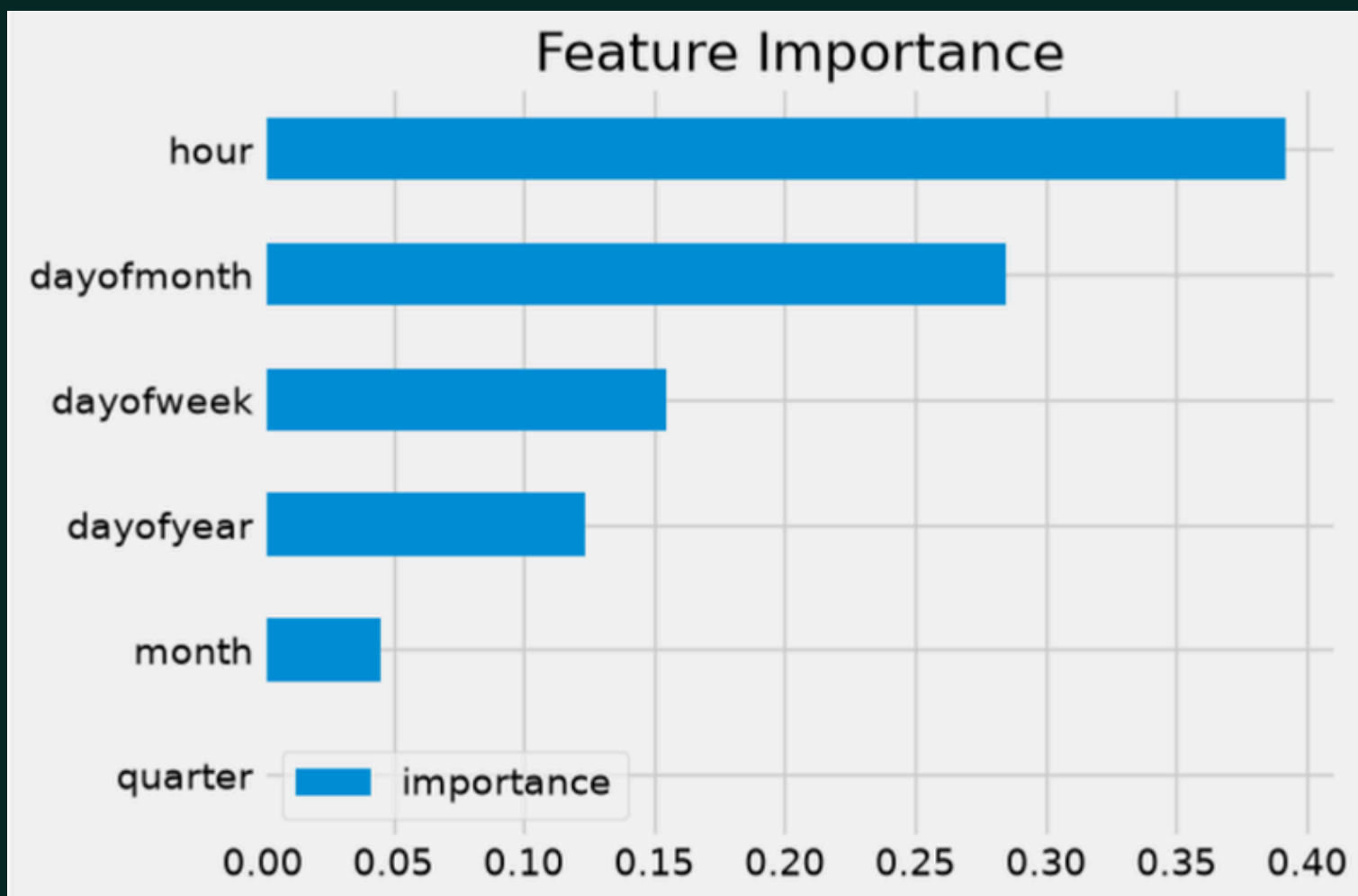
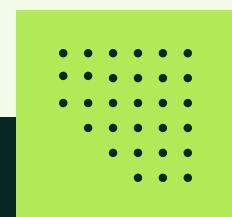
ML Model Features

● ● ● What Drives Energy Consumption?

Model Insights: The "Why" Behind the Forecast

The XGBoost model identified Hour(39%) and DayofMonth (28%) as the most important factors, confirming the strong daily and weekly cycles observed in the data.

Key Takeaway: The model's logic aligns with real-world intuition, building trust in its predictions.



Feature Importance

● ● ● Putting the Model to the Test

Error Analysis

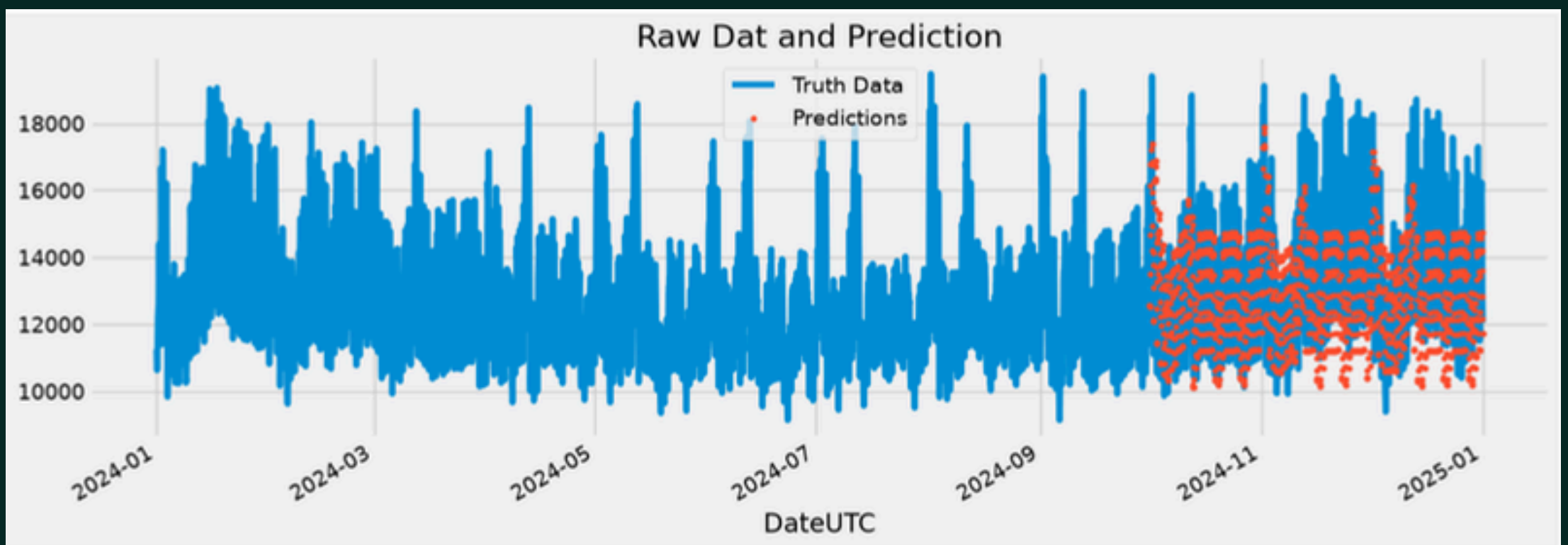
Best days: early Oct (317–395 MW error).

Worst days: 23 Nov, 14 Dec, 20 Nov, 13 Dec, 22 Nov (2,496–2,790 MW error).

Likely weather/holiday effects.

Data Callout: Max daily mean error = 2,790 MW

Key Takeaway: The model provides a strong foundational forecast with a quantifiable level of accuracy.



Raw Dat and Prediction



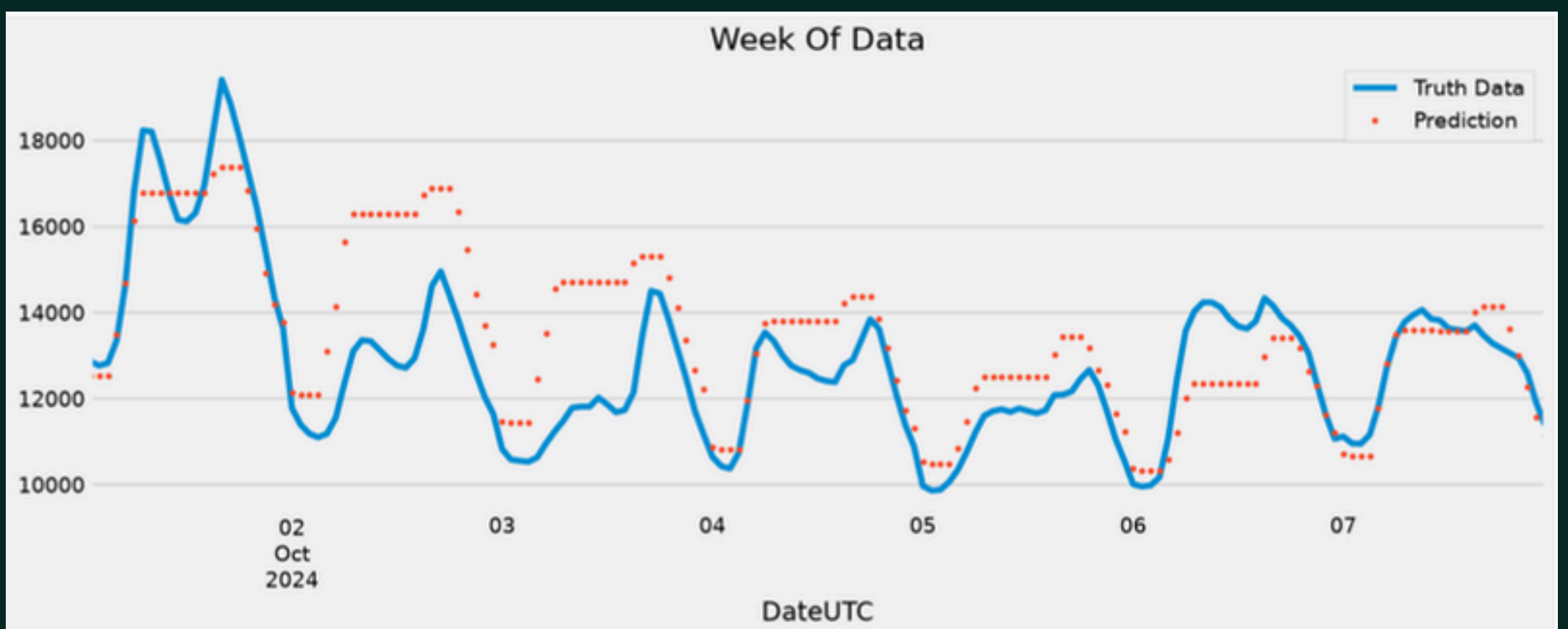
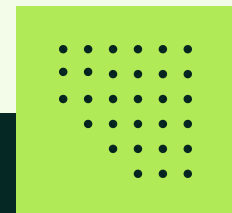
ML Model Testing

● ● ● A Closer Look at Performance

Truth vs. Prediction: A Week in Review

The model successfully captures the overall shape of daily demand but struggles with the exact magnitude of some peaks, indicating an area for future improvement.

Key Takeaway: Visual inspection confirms the model tracks trends well, a key requirement for practical use.



Week Of Data



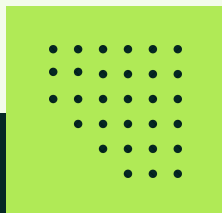
● ● ● Learning from Mistakes

When the Model Gets It Wrong

Best days: early Oct (317–395 MW error).

Worst days: 23 Nov, 14 Dec, 20 Nov, 13 Dec, 22 Nov (2,496–2,790 MW error).

Key Takeaway: Add binary holiday flags and temperature data to reduce large errors during anomalous periods.



date	
2024-11-23	2790.234295
2024-12-14	2733.290155
2024-11-20	2603.502277
2024-12-13	2547.517352
2024-11-22	2496.552144

Worst Predicted Days

date	
2024-10-07	317.421165
2024-10-26	362.230028
2024-11-04	371.455815
2024-10-27	378.322427
2024-10-10	394.567992

Best Predicted Days

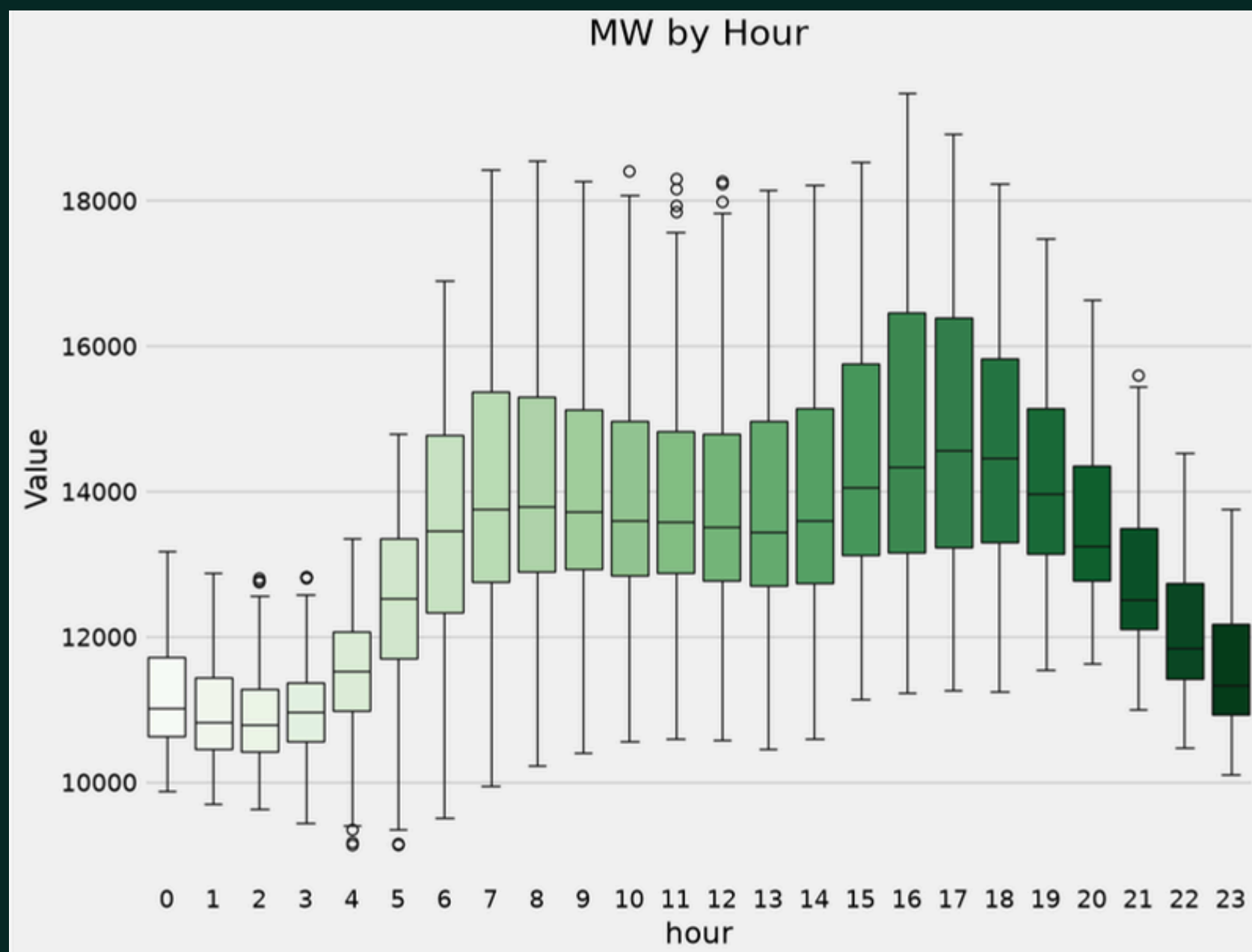
Strategic Recommendations

1. Add exogenous variables: temperature, holidays, daylight.
2. Add lags: 1h, 24h, 168h; rolling stats.
3. Use cyclical encodings (sin/cos).
4. Remove redundant calendar features.
5. Use time-series cross-validation.
6. Compare with SARIMA, LightGBM, LSTM.

 Mohamed Tawhid

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Call to Action



MW by Hour